Chapter 12
Partial Least Squares Path Modeling

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12.1 Introduction

Structural equation modeling (SEM) is a family of statistical techniques that has become very popular in marketing. Its ability to model latent variables, to take various forms of measurement error into account, and to test entire theories makes it useful for a plethora of research questions. It does not come as a surprise that some of the most cited scholarly articles in the marketing domain are about SEM (e.g., Bagozzi and Yi 1988; Fornell and Larcker 1981), and that SEM is covered by two contributions within this volume. The need for two contributions arises from the SEM family tree having two major branches (Reinartz et al. 2009): covariance-based SEM (which is presented in Chap. 11) and variance-based SEM, which is presented in this chapter.

Covariance-based SEM estimates model parameters using the empirical variance–covariance matrix, and is the method of choice if the hypothesized model consists of one or more common factors. In contrast, variance-based SEM first creates proxies as linear combinations of observed variables, and then estimates the model parameters using these proxies. Variance-based SEM is the method of choice if the hypothesized model contains composites. Of the variance-based SEM methods, partial least squares path modeling (PLS) is regarded as the "most fully developed and general system" (McDonald 1996, p. 240), and is the subject of this contribution.

A distinguishing PLS characteristic is its ability to include both factors and composites in a structural equation model (Dijkstra and Henseler 2015a, 2015b). Factors can be used to model latent variables of behavioral research, such as

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attitudes or personality traits. Another term for this type of model is reflective measurement. Composites can be applied to model abstractions of artifacts such as plans, strategies, value, portfolios, and marketing instruments in general (Henseler 2015). In this vein, Albers (2010) recommends PLS as the preferred statistical tool for success factor studies in marketing. Composites are the core of composite–formative measurement (Bollen and Diamantopoulos 2017).

Recently, PLS has undergone a series of serious examinations, and has been the topic of heated scientific debates (Henseler et al. 2016). Scholars have discussed the conceptual underpinnings (Rigdon 2012, 2014; Sarstedt et al. 2014), the strengths and weaknesses (Henseler et al. 2014; Rigdon et al. 2014), and the use of PLS as a statistical method (Hair et al. 2012a, 2012b). As a fruitful outcome of these debates, substantial contributions to PLS emerged, such as bootstrap-based tests of the overall model fit (Dijkstra and Henseler 2015a), consistent PLS with which to estimate factor models (PLSc, see Dijkstra and Henseler 2015b), and the heterotrait–monotrait ratio of correlations as a new criterion for discriminant validity (HTMT, see Henseler et al. 2015). These new developments call for updated guidelines on why, when, and how to use PLS in marketing research.

The purpose of this chapter is manifold. In Sect. 12.2, it provides an updated view on what PLS actually is and the algorithmic steps it has included since the invention of consistent PLS. In Sect. 12.3, it explains how to specify PLS path models, taking the nature of the measurement models (composite vs. factor), model identification, sign indeterminacy, and special treatments of categorical variables into account. In Sect. 12.4, it explains how to assess and report PLS results, including the novel bootstrap-based tests of model fit, the SRMR as an approximate measure of model fit, the new reliability coefficient rho_A, and the HTMT. In Sect. 12.5, we discuss the various publicly available software implementations. An example application of PLS to illustrate its use is given in Sect. 12.6. Finally, Sect. 12.7 provides a concluding discussion.

12.2 The Partial Least Squares Path Modeling Method

The core of PLS is a set of alternating least squares algorithms that emulates and extends principal component analysis, as well as canonical correlation analysis. Herman Wold (1974, 1982) invented the method, which has undergone various extensions and modifications, to analyze high dimensional data in a low-structure environment. In its most modern appearance (see Dijkstra and Henseler 2015a, 2015b), PLS path modeling can be understood as a full-fledged structural equation modeling method that can handle both factor models and composite models for construct measurement, can estimate recursive and non-recursive structural models, and conduct exact tests of model fit.

PLS path models are formally defined by two sets of linear equations: the measurement model (also called the outer model) and the structural model (also called the inner model). The measurement model specifies the relations between

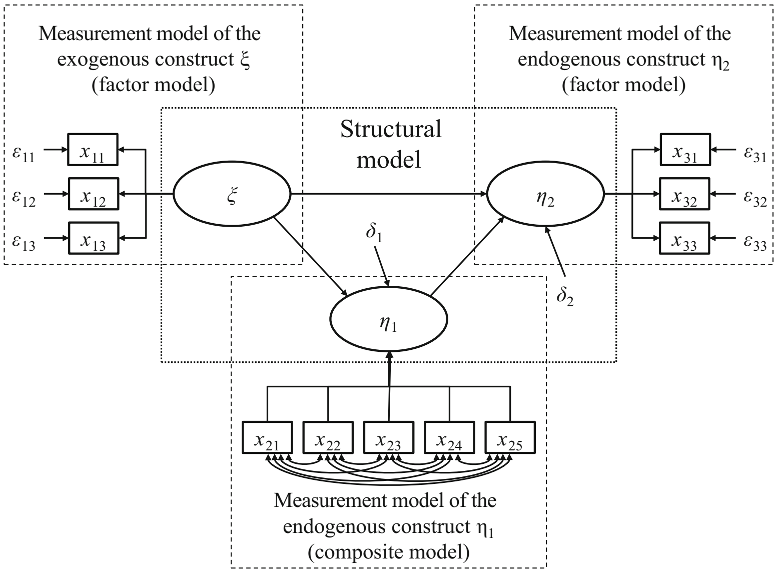


Fig. 12.1 PLS path model example

a construct and its observed indicators (also called manifest variables), whereas the structural model specifies the relationships between the constructs. Figure 12.1 depicts an example of a PLS path model.

The structural model consists of exogenous and endogenous constructs, as well as the relationships between them. The values of exogenous constructs are assumed to be given from outside the model. Thus, other constructs in the model do not explain exogenous variables, and no arrows in the structural model should point to exogenous constructs. In contrast, other constructs in the model explain endogenous constructs at least partially. Each endogenous construct must have at least one of the structural model's arrow pointing to it. The relationships between the constructs are usually assumed to be linear. The size and significance of path relationships are typically the focus of the scientific endeavors pursued in empirical research.

PLS path models can contain two different forms of construct measurement: factor models and composite models (see Rigdon 2012, for a nice comparison of both types of measurement models). The factor model hypothesizes that the existence of one unobserved variable (the common factor) and indicator-specific random error perfectly explain the variance of a block of indicators. This is the standard model of behavioral research. The term reflective measurement model is also often used. In Fig. 12.1, the exogenous construct ξ and the endogenous construct η_2 are modeled as factors. In contrast, composites are formed as linear combinations of their respective indicators. The composite model does not impose any restrictions on the covariances between indicators of the same construct, i.e., it relaxes the assumption that a common factor explains all the covariation between a block of indicators.

The estimation of PLS path model parameters is done in four steps: An iterative algorithm that determines the composite scores of each construct, a correction for attenuation of those constructs modeled as factors (Dijkstra and Henseler 2015b), parameter estimation, and bootstrapping for inference testing.

Step 1 For each construct, the iterative PLS algorithm creates a proxy as a linear combination of the observed indicators. The indicator weights are determined such that each proxy shares as much variance as possible with the proxies of causally related constructs. The PLS algorithm can be viewed as an approach to extend canonical correlation analysis to more than two sets of variables; it can emulate several of Kettenring's (1971) techniques for the canonical analysis of several sets of variables (Tenenhaus et al. 2005). For a more detailed description of the algorithm, see Henseler (2010). The proxies (i.e., composite scores), the proxy correlation matrix, and the indicator weights are the major output of the first step.

Step 2 Correcting for attenuation is a necessary step if a model involves factors. As long as the indicators contain random measurement error, so will the proxies. Consequently, proxy correlations are usually underestimations of factor correlations. Consistent PLS (PLSc) corrects this tendency (Dijkstra and Henseler 2015a, 2015b) by dividing a proxy's correlations with other proxies by the square root of its reliability (also known as the correction for attenuation). PLSc addresses the issue of what the correlation between constructs would be if there were no random measurement error. The major output of this second step is a consistent construct correlation matrix.

Step 3 Once a consistent construct correlation matrix is available, it is possible to estimate the model parameters. If the structural model is recursive (i.e., there are no feedback loops), ordinary least squares (OLS) regression can be used to obtain consistent parameter estimates of the structural paths. In the case of non-recursive models, instrumental variable techniques, such as two-stage least squares (2SLS), should be employed. Beside the path coefficient estimates, this third step can also provide estimates of loadings, indirect effects, total effects, and several model assessment criteria.

Step 4 Finally, the bootstrap is applied in order to obtain inference statistics for all the model parameters. The bootstrap is a non-parametric inferential technique based on the assumption that the sample distribution conveys information about the population distribution. Bootstrapping is the process of drawing a large number of re-samples with replacement from the original sample, and then estimating the model parameters for each bootstrap re-sample. The standard error of an estimate is inferred from the standard deviation of the bootstrap estimates.

The PLS path modeling algorithm has favorable convergence properties (Henseler 2010). However, as soon as PLS path models involve common factors, there is the possibility of Heywood cases (Krijnen et al. 1998), meaning that one or more of the variances that the model implies could be negative. An atypical, or too-small sample, may cause the occurrence of Heywood cases, or the common factor structure may not hold for a particular block of indicators.

PLS path modeling is not as efficient as maximum likelihood covariance-based structural equation modeling. One possibility is to further minimize the discrepancy between the empirical and the model-implied correlation matrix, an approach that efficient PLS follows (PLSe, see Bentler and Huang 2014). Alternatively, one could embrace the notion that PLS is a limited-information estimator and that model misspecification in some subparts of a model affects it less (Antonakis et al. 2010). Ultimately, there is no clear-cut resolution of the issues on this trade-off between efficiency and robustness with respect to model misspecification.

12.3 Specifying PLS Path Models

The analysts must ensure that the specified statistical model complies with the conceptual model intended to be tested, and further, that the model complies with the technical requirements such as identification, and with the data conforming to the required format and offering sufficient statistical power.

Typically, the structural model is theory-based and is the prime focus of the research question and/or research hypotheses. The specification of the structural model addresses two questions: Which constructs should be included in the model? And how are they hypothesized to be interrelated? That is, what are the directions and strengths of the causal influences between and among the constructs? In general, analysts should keep in mind that the constructs specified in a model are only proxies, and that there will always be a validity gap between these proxies and the theoretical concepts that are the intended modeling target (Rigdon 2012). The paths, specified as arrows in a PLS model, represent directional linear relationships between these proxies. The structural model and the indicated relationships among the constructs are regarded as separate from the measurement model.

The specification of the measurement model entails deciding on composite or factor models and assigning indicators to constructs. Factor models are the predominant measurement model for behavioral constructs such as attitudes or personality traits. Factor models are strongly linked to true score theory (McDonald 1999), the most important measurement paradigm in behavioral sciences. If a construct has this background and random measurement error is likely to be an issue, analysts should choose the factor model. In contrast composites help model emergent constructs, for which elements are combined to form a new entity (Henseler 2017). Composites can be applied to model strong concepts (Höök and Löwgren 2012), i.e., the abstraction of artifacts. Whenever a model contains this type of construct, it is preferable to opt for a composite model.

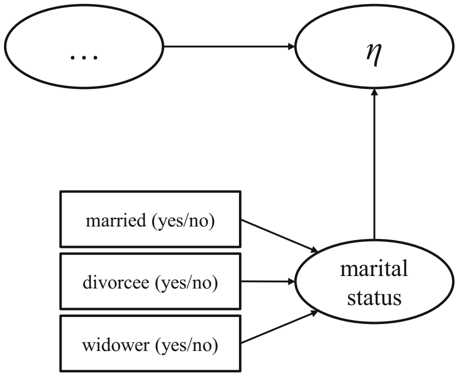
Measurement models of PLS path models may appear less detailed than those of covariance-based structural equation modeling, but some specifications are implicit and not visualized. For instance, neither the unique indicator errors (nor their correlations) of factor models, nor the correlations between the indicators of composite models are drawn. Structural disturbance terms are assumed to be orthogonal to their predictor variables, as well as to each other[1]; and correlations between exogenous variables are free. Because PLS does not currently allow either constraining these parameters, or freeing the error correlations of factor models, these model elements are, by convention, not drawn. No matter which type of measurement is chosen to measure a construct, PLS requires at least one indicator. Constructs without indicators, also called phantom variables (Rindskopf 1984), cannot be included in PLS path models.

Identification has always been an important issue for SEM, although it was previously neglected in the realm of PLS path modeling. It refers to the necessity to specify a model such that only one set of estimates exists that yields the same model-implied correlation matrix. A complete model might be unidentified, as might only parts of it. In general, it is not possible to derive useful conclusions from unidentified (parts of) models. In order to achieve identification, PLS fixes the variance of factors and composites to one. A nomological net is an important requirement of composite models. This means that composites cannot be estimated in isolation, but need at least one relation with another variable. Since PLS also estimates factor models via composites, this requirement extends to all factor models estimated by using PLS. If a factor model has exactly two indicators, it does not matter which form of measurement model is used—a nomological net is then required to achieve identification. If only one indicator measures a construct, this is called a single-indicator measurement (Diamantopoulos et al. 2012). The construct scores are then identical to the standardized indicator values. In this case, the amount of random measurement error in this indicator cannot be determined. If an indicator contains measurement error, the only possibility to account for the error is to utilize external knowledge about this indicator's reliability and to manually define this.

A typical characteristic of SEM, and factor-analytical tools in general, is sign-indeterminacy, in which the weight, or loading estimates, of a factor or a composite can only be unanimously determined regarding their value, but not regarding their sign. For example, if a factor is extracted from the strongly negatively correlated customer satisfaction indicators "How satisfied are you with provider X?" and "How much does provider X differ from an ideal provider?," the method cannot "know" whether the extracted factor should correlate positively with the first or with the second indicator. Depending on the sign of the loadings, the meaning of the factor would either be "customer satisfaction" or "customer non-satisfaction." To avoid this ambiguity, it has become practice in SEM to determine a particular indicator per construct, with which the construct scores are then forced to correlate positively.

[1]This assumption should be relaxed in the case of non-recursive models (Dijkstra and Henseler 2015a).

Fig. 12.2 Including a categorical variable in a PLS path model
Note: Marital status with the four categories "unmarried," "married," "divorcee," "widower"; the reference category is "unmarried"



Since this indicator dictates the orientation of the construct, it is called the dominant indicator. Whereas, in covariance-based structural equation modeling, this dominant indicator also dictates the construct's variance, in PLS path modeling the construct variance is simply set to one.

Like multiple regression, PLS path modeling requires metric data for the dependent variables. Dependent variables are the indicators of the factor model(s), as well as the endogenous constructs. Quasi-metric data stemming from multi-point scales, such as Likert scales or semantic differential scales, are also acceptable as long as the scale points can be assumed to be equidistant and there are five or more scale points (Rhemtulla et al. 2012). To some extent it is also possible to include categorical variables in a model. Categorical variables are particularly relevant for analyzing experiments (see Streukens et al. 2010), or for control variables. Figure 12.2 illustrates how a categorical variable "marital status" would be included in a PLS path model. If a categorical variable has only two levels (i.e., it is dichotomous), it can immediately serve as a construct indicator. If a categorical variable has more than two levels, it should be transformed into as many dummy variables as there are levels. A composite model is built from all but one dummy variable. The remaining dummy variable characterizes the reference level. Preferably, categorical variables should only play the role of exogenous variables in a structural model.

12.4 Assessing and Reporting PLS Analyses

PLS path modeling can be used for both explanatory and predictive research. The model assessment will differ depending on the analyst's aim—explanation or prediction. Since PLS applications in marketing mainly focus on explanation (Hair et al. 2012b), in the remainder, we concentrate on model assessment given that the analyst's aim is explanation.

12.4.1 Assessing Overall Model Fit

PLS results can be assessed globally (i.e., for the overall model) and locally (separately for the measurement model and the structural model). Since, in the form described above, PLS provides consistent estimates for factor and composite models, it is possible to meaningfully compare the model-implied correlation matrix with the empirical correlation matrix, which opens up the possibility to assess the global model fit.

The model's overall goodness of fit should be the starting point of model assessment. If the model does not fit the data, the data contain more information than the model conveys. The obtained estimates may be meaningless, in which case the conclusions drawn from them become questionable. The global model fit can be assessed in two non-exclusive ways: by means of inference statistics, i.e., tests of the model fit, or through the use of fit indices, i.e., an assessment of the approximate model fit. In order to have some frame of reference, it has become customary to determine the model fit for both the estimated model and the saturated model. Saturation refers to the structural model, which means that, in the saturated model, all the constructs correlate freely. Any lack of fit of the saturated model can only be attributed to the construct measurement. Hence, the saturated model is most suitable for assessing the measurement model, whereas the estimated model also allows to quantify the fit of the structural model.

Tests of the overall model fit of PLS path models rely on the bootstrap to determine the likelihood of obtaining a discrepancy between the empirical and the model-implied correlation matrix that is as high as the that obtained for the sample at hand, if the hypothesized model were indeed correct (Dijkstra and Henseler 2015a). Bootstrap samples are drawn from modified sample data. This modification entails an orthogonalization of all variables and a subsequent imposition of the model-implied correlation matrix. In covariance-based SEM, this approach is known as the Bollen–Stine bootstrap (Bollen and Stine 1992). If more than five percent (or a different percentage if an alpha level different from 0.05 is chosen) of the bootstrap samples yield discrepancy values above the ones of the actual model, the sample data are likely to stem from a population that functions according to the hypothesized model. The model thus cannot be rejected.

There is more than one way to quantify the discrepancy between two matrices, for instance, the maximum likelihood discrepancy, the geodesic discrepancy d_G, or the unweighted least squares discrepancy d_{ULS} (Dijkstra and Henseler 2015a), and there are consequently several tests of model fit. Monte Carlo simulations confirm that the tests of model fit can indeed discriminate between well-fitting and ill-fitting models (Henseler et al. 2014). More precisely, both measurement model misspecification and structural model misspecification can be detected by testing the model fit (Dijkstra and Henseler 2014). Since different tests might lead to different results, a transparent reporting practice should always include several tests.

Beside conducting model fit tests, the approximate model fit can also be determined. Approximate model fit criteria help answer the question of how substantial the discrepancy between the model-implied and the empirical correlation matrix is. This question is particularly relevant if this discrepancy is significant, or if a too small sample size and the subsequently low statistical power renders the model fit tests too liberal. Currently, in the context of PLS, the dominant approximate model fit criterion is the standardized root mean square residual (SRMR, Hu and Bentler 1998, 1999). As can be derived from its name, the SRMR is the square root of the sum of the squared differences between the model-implied and the empirical correlation matrix, i.e., the Euclidean distance between the two matrices. A value of 0 for the SRMR would indicate a perfect fit, and generally an SRMR value less than 0.05 indicates an acceptable fit (Byrne 2013). However, even entirely correctly specified PLS path models can yield SRMR values of 0.06 and higher (Henseler et al. 2014). Therefore, a cut-off value of 0.08, which Hu and Bentler (1999) propose, appears to be more adequate for PLS. Another useful approximate model fit criterion could be the Bentler–Bonett index, or the normed fit index (NFI, Bentler and Bonett 1980), which Lohmöller (1989) suggested using in connection with PLS path modeling. NFI values above 0.90 are considered acceptable for factor models (Byrne 2013). Thresholds for the NFI are still to be determined regarding composite models. Further, the NFI does not penalize the adding of parameters and should thus be used with caution for model comparisons. In general, the usage of the NFI is still rare.[2] The root mean square error correlation ($\mathrm{RMS}_{\mathrm{theta}}$, see Lohmöller 1989) is another promising approximate model fit criterion. While the $\mathrm{RMS}_{\mathrm{theta}}$ can distinguish well-specified from ill-specified models (Henseler et al. 2014), the $\mathrm{RMS}_{\mathrm{theta}}$ thresholds are yet to be determined, and PLS software has not yet implemented this approximate model fit criterion. Note that early suggestions for PLS-based goodness-of-fit measures, such as the “goodness-of-fit” (GoF, see Tenenhaus et al. 2004) or the “relative goodness-of-fit” ($\mathrm{GoF}_{\mathrm{rel}}$, proposed by Esposito Vinzi et al. 2010), are—contrary to what they seem to suggest—not informative about the goodness of the model fit (Henseler et al. 2014; Henseler and Sarstedt 2013). Consequently, there is no reason to evaluate and report them if the analyst’s aim is to test or compare models.

12.4.2 Assessing Measurement Models

If the specified measurement (or outer) model does not have the minimum required properties of acceptable reliability and validity, the structural (inner) model estimates become meaningless. That is, a necessary condition before starting to assess the “goodness” of the inner structural model is that the outer measurement model should already demonstrate acceptable levels of reliability and validity. There must be a sound measurement model before one can begin to assess the “goodness” of the inner structural model, or can rely on the magnitude, direction, and/or statistical

[2]For an application of the NFI, see Ziggers and Henseler (2016).

strength of the structural model's estimated parameters. Factor and composite models are assessed differently.

Factor models can be assessed in various ways. The bootstrap-based tests of overall model fit (of the saturated model) can indicate whether the data are coherent with a factor model, i.e., it represents a confirmatory factor analysis. In essence, the test of model fit provides an answer to the question "Does empirical evidence negate the existence of the factor?" This quest for truth illustrates that the factor model testing is rooted in the positivist research paradigm. If the overall model fit test does not provide evidence negating the existence of a factor,[3] several questions regarding the factor structure emerge: Do the data support a factor structure at all? Can one clear factor be consistently extracted? How well has this factor been measured? Note that tests of overall model fit cannot answer these questions; specifically, entirely uncorrelated empirical variables do not necessarily lead to the factor model's rejection. To answer these questions, one should instead rely on various local assessment criteria regarding the reliability and validity of measurement.

The amount of random error in the construct scores should be acceptable; that is, the reliability of the construct scores should be sufficiently high. Nunnally and Bernstein (1994) recommend a minimum reliability of 0.7. The most important PLS reliability measure is rho_A (Dijkstra and Henseler 2015b), which is currently the only consistent reliability measure of PLS construct scores. The reliability measure rho_A is an estimate for the squared correlation of the PLS construct score with the (unknown) true construct score. Most PLS software also provides a measure of composite reliability (also called Dillon-Goldstein's rho, factor reliability, Jöreskog's rho, omega, or rho_c), as well as Cronbach's alpha. Both refer to sum scores, not composite scores. In particular, Cronbach's alpha typically underestimates the true reliability, and should therefore only be regarded as a lower boundary of the reliability (Sijtsma 2009).

The measurement of factors should also be free of systematic measurement error. This quest for validity can be fulfilled in several non-exclusive ways. First, a factor should be unidimensional, a characteristic that convergent validity examines. The dominant measure of convergent validity is the average variance extracted (AVE, Fornell and Larcker 1981).[4] If the first factor extracted from a set of indicators explains more than one half of their variance, there cannot be a second, equally important, factor. An AVE of 0.5 or higher is therefore regarded as acceptable. Sahmer et al. (2006) proposed a somewhat more liberal criterion: They find evidence of unidimensionality as long as a factor explains significantly more variance than the second factor extracted from the same block of indicators. Second, each pair

[3]Interestingly, the methodological literature on factor models hardly mentions what to do if the test rejects a factor model. Some researchers suggest considering a composite model as an alternative, because it is less restrictive (Henseler et al. 2014) and not subject to factor indeterminacy (Rigdon 2012). Others suggest allowing small deviations without principally questioning the factor model (see Asparouhov et al. 2015).

[4]The AVE must be calculated based on consistent loadings, otherwise the assessment of convergent and discriminant validity based on the AVE is meaningless.

of factors that represent theoretically different concepts should also be statistically different, which raises the question of discriminant validity. Two criteria have been shown to be informative about discriminant validity (Voorhees et al. 2016): The Fornell–Larcker criterion (proposed by Fornell and Larcker 1981) and the heterotrait–monotrait ratio of correlations (HTMT, developed by Henseler et al. 2015). The Fornell-Larcker criterion maintains that a factor's AVE should be higher than its squared correlations with all other factors in the model. The HTMT is an estimate of the factor correlation (more precisely, an upper boundary). In order to clearly discriminate between two factors, the HTMT should be significantly smaller than one. Third, the cross-loadings should be assessed to ensure that no indicator is incorrectly assigned.

The assessment of composite models is somewhat less developed. Again, the major point of departure should be the tests of model fit. The tests of the model fit of the saturated model provide evidence of the composites' external validity. Henseler et al. (2014) call this step a "confirmatory composite analysis." For composite models, the major research question is "Does it make sense to create this composite?" This question shows that testing composite models follows a different research paradigm, namely pragmatism (Henseler 2015). Once confirmatory composite analysis has provided support for the composite, it can be analyzed further. Some follow-up questions present themselves: How is the composite built? Do all the ingredients contribute significantly and substantially? To answer these questions, an analyst should assess the sign and the magnitude of the indicator weights, as well as their significance. If indicator weights have unexpected signs, or are insignificant, this can specifically be due to multicollinearity. It is therefore recommendable to assess the variance inflation factor (VIF) of the indicators. VIF values far higher than one indicate that multicollinearity might play a role. In this case, analysts should consider using correlation weights (PLS Mode A, see Rigdon 2012), or the best fitting proper indices (Dijkstra and Henseler 2011) to estimate the indicator weights.

12.4.3 Assessing Structural Models

Once the measurement model is deemed of sufficient quality, the analyst can proceed and assess the structural model. If OLS is used for the structural model, the endogenous constructs' R^2 values would be the point of departure. They indicate the percentage of variability accounted for by the precursor constructs in the model. The adjusted R^2 values take the model complexity and sample size into account, and are thus helpful to compare different models, or the explanatory power of a model across different datasets.

The path coefficients are essentially standardized regression coefficients, which can be assessed with regard to their sign and their absolute size. They should be interpreted as the change in the dependent variable if the independent variable is increased by one and all other independent variables remain constant. Indirect effects and their inference statistics are important for mediation analysis (Nitzl et al. 2016; Zhao et al. 2010), while total effects are useful for successful factor analysis (Albers 2010).

If the analyst's aim is to generalize from a sample to a population, the path coefficients should be evaluated for significance. Inference statistics include the empirical bootstrap confidence intervals, as well as one-sided or two-sided p-values. We recommend using 4999 bootstrap samples. This number is sufficiently close to infinity for usual situations, is tractable with regard to computation time, and allows for a unanimous determination of empirical bootstrap confidence intervals (for instance, the 2.5% [97.5%] quantile would be the 125$^{\text{th}}$ [4875$^{\text{th}}$] element of the sorted list of bootstrap values). A path coefficient is regarded as significant (i.e., unlikely to purely result from sampling error) if its confidence interval does not include the value of zero, or if the p-value is below the pre-defined alpha level. Despite strong pleas for the use of confidence intervals (Cohen 1994), reporting p-values still seems to be more common in business research.

It makes sense to quantify how substantial the significant effects are, which can be done by assessing their effect size f^2. Effect size values above 0.35, 0.15, and 0.02 can be regarded as respectively strong, moderate, and weak (Cohen 1988).

Finally, recent research confirms that PLS is a promising technique for prediction purposes (Becker et al. 2013). Blindfolding is the standard approach used to examine if the model, or a single effect of it, can predict the values of reflective indicators (Tenenhaus et al. 2005). It is already widely applied (Hair et al. 2012b; Ringle et al. 2012). Criteria for the predictive capability of structural models have been proposed (see Chin 2010), but still need to be disseminated. Once business and social science researchers' interest in prediction becomes more pronounced, PLS is likely to face an additional substantial increase in popularity.

12.5 PLS Software

There is quite a variety of PLS software available, each of which has unique advantages and disadvantages. The first widely available PLS software was LVPLS (Lohmöller 1988), which did not yet contain a graphical user interface. The further development of the program was discontinued after the early death of the program author. Since the original code is no longer available, changes to the calculation of LVPLS are hardly possible. However, Wynne Chin substantially increased the usability of LVPLS by embedding it into a graphical user interface called PLS-Graph (Chin and Frye 2003). PLS-Graph was the dominant software at the end of the last millennium. Its limited improvement and extension possibilities motivated Christian Ringle and his team to develop a new PLS software, called SmartPLS, from scratch (Ringle et al. 2005). Later, other PLS applications emerged, such as PLS-GUI, WarpPLS, and XLSTAT-PLS.

When specifying the model, analysts should keep in mind that, in some PLS path modeling software (e.g., SmartPLS, PLS-Graph, and XLSTAT-PLS), the depicted direction of the arrows in the measurement model does not indicate whether a factor or composite model is estimated. Instead, the arrow directions indicate whether correlation weights (Mode A, represented by arrows pointing from a construct to its indicators) or regression weights (Mode B, represented by arrows pointing from indicators to their construct) should be used to create the proxy. Mode A uses a set of simple regressions to determine the indicator weights, whereas Mode B uses a multiple regression. PLS will estimate a composite model in both cases. Indicator weights estimated by Mode B are consistent (Dijkstra 2010), whereas indicator weights estimated by Mode A are usually not consistent. However, the latter excel at out-of-sample prediction (Rigdon 2012).

Of all the PLS programs with graphical user interface, SmartPLS 3.2 (Ringle et al. 2015) is currently the most comprehensive software. It contains many extensions of PLS, such as analysis of interaction effects (Henseler and Chin 2010; Henseler and Fassott 2010), analysis of nonlinear effects (Henseler et al. 2012a), multigroup analysis (Henseler 2012a; Sarstedt et al. 2011), assessment of measurement invariance (Jean et al. 2016), importance-performance matrix analysis (Ringle and Sarstedt 2016), and diagnostics for predictive research like blindfolding (Tenenhaus et al. 2005). However, if an analyst undertakes confirmatory research, SmartPLS is not optimally suitable, because the model fit tests are not implemented (version 3.2). Analysts may prefer ADANCO (Henseler and Dijkstra 2015), a new software for variance-based SEM, which also includes PLS path modeling. ADANCO has implemented all goodness-of-fit criteria presented in this contribution, including the tests of the overall model fit.

12.6 Empirical Application

A researcher wants to explore whether there is a relationship between customer focus and firm performance. The researcher has empirical data from 176 key informants. These data include six reflective indicators of the customer focus; measures of return on investment, the profit margin, the profit, and the market capitalization; and a categorical variable capturing the industry.

Figure 12.3 depicts the model as specified, using ADANCO 2.0. It consists of the endogenous construct firm performance as a composite of return on investment, the profit margin, the profit, and the market capitalization; the exogenous construct customer focus measured by the six reflective indicators; and a control variable industry composed of a set of dummy variables like the set proposed in Fig. 12.2. Figure 12.3 also shows the most relevant model estimates: path coefficients (and their significance), weights (of the composite models), and loadings (of the factor models).

The assessment of the construct measurement focuses on two major questions: Can we clearly extract one factor from our six customer focus indicators? And is it reasonable to create a firm performance composite as a linear combination of return on investment, the profit margin, the profit, and the market capitalization? The

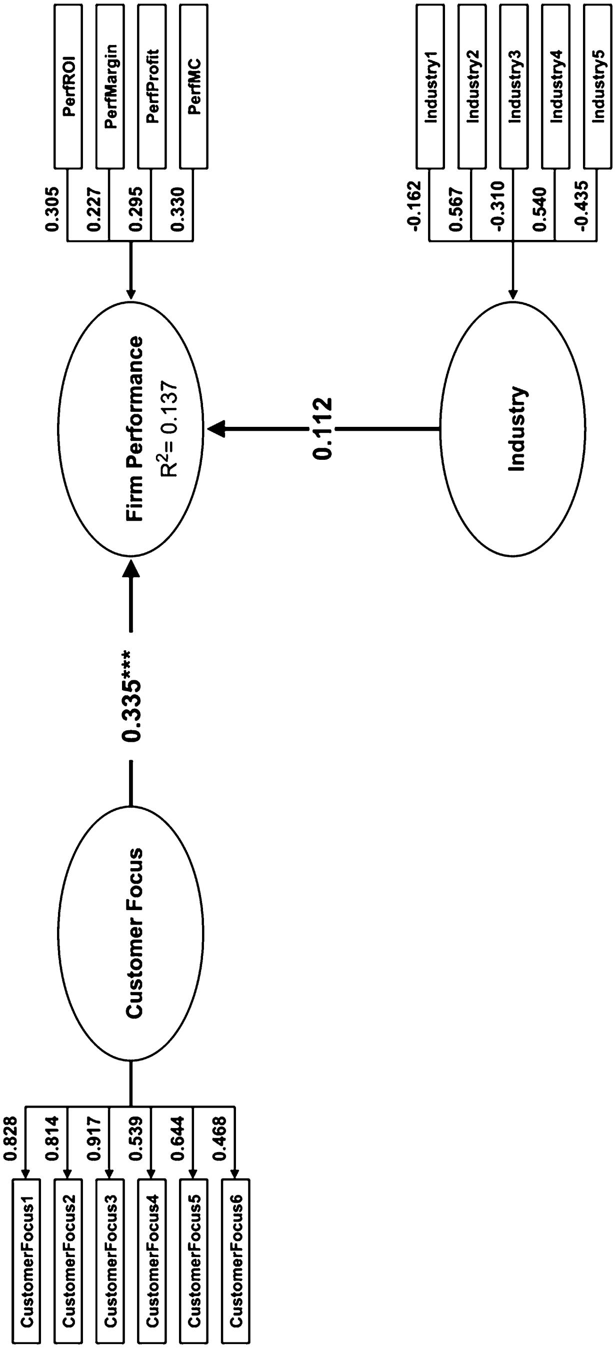


Fig. 12.3 Example model as specified, using ADANCO 2.0

overall model fit tests of the saturated model simultaneously conduct confirmatory factor analysis (to answer the first question) and confirmatory composite analysis (to answer the second question). ADANCO provides three measures of discrepancy between the empirical and the model-implied correlation matrix, together with the 95% quantile of its distribution if the model is correct (HI_{95}): an SRMR value of 0.069 ($HI_{95} = 0.100$), a d_{ULS} value of 0.575 ($HI_{95} = 1.192$), and a d_G value of 0.285 ($HI_{95} = 0.586$). All three measures of discrepancy are below their corresponding HI_{95} values, which means that the discrepancy between the empirical and the model-implied correlation matrix is not significant. This implies that the information loss owing to the composite of firm success is negligible, and can be defended to form this composite.

Since the customer focus construct is operationalized using a factor model, we can apply the criteria for reliability and construct validity. The reliability coefficient ρ_A is 0.890, which implies a high degree of reliability. This is higher than ρ_c (0.860) and α (0.869), which means that it is worth the effort to create the construct scores as a weighted sum of its indicators instead of pure sum scores. The average variance extracted is 0.519, which provides evidence of the unidimensionality of the customer focus construct. Since customer focus is the only construct operationalized using a factor model, neither the Fornell-Larcker criterion, nor the HTMT can be applied to assess discriminant validity. However, given the inter-construct correlations below 0.4, there is hardly any basis for doubts about the discriminant validity. Overall, the construct measurement can be deemed valid.

Once we have sufficient certainty about the quality of the construct measurement, we can assess the structural model. Because the structural model is saturated, the overall model fit of the estimated model equals that of the saturated model. This implies that the overall model fit does not inform about the structural model. However, this would only be a problem if the researcher's aim is confirmatory research. A local assessment of the structural model provides sufficient insight regarding exploratory research. First, we look at the coefficient of determination (R^2 value) of the endogenous variable, namely firm performance. While the obtained R^2 value of 0.137 may not be large, it is certainly worthwhile interpreting this value relatively to the R^2 values obtained in comparable studies, because normally achieved R^2 values tend to vary across disciplines and phenomena.

Inference statistics based on 4999 bootstrap samples indicate that the effect of customer focus on firm performance is significant ($p < 0.001$), whereas firm performance does not vary significantly across industries. The f^2 value of the effect of customer focus on firm performance is 0.127, which means that its effect size is small to moderate. The path coefficient of 0.335 means that if one of two firms in the same industry succeeds in increasing its customer focus by one standard deviation, it will gain an increase in firm performance of 0.335 standard deviations.

12.7 Further Applications and Outlook

Traditionally, national customer satisfaction indices have been the dominant field of PLS path model application in marketing. From the beginning, PLS path modeling has been the method of choice of the Swedish Customer Satisfaction Index (Fornell 1992), and continues to be the method of choice for successors, such as the American Customer Satisfaction Index (Fornell et al. 1996), the European Customer Satisfaction Index (Tenenhaus et al. 2005), and the Portuguese Customer Satisfaction Index (Coelho and Henseler 2012). However, the use of PLS is in no way limited to customer satisfaction indices. Hair et al. (2012b) identified hundreds of PLS path models reported in the leading marketing journals. Moreover, PLS is applied in various marketing subdisciplines, such as advertising (Henseler et al. 2012b) and international marketing (Henseler et al. 2009). Some studies are particularly worth naming:

- Rego (1998) reports the probably smallest PLS path model in the marketing literature. It consists of two constructs, market structure and market efficiency. This model provides evidence for both a negative linear and a negative quadratic effect of market structure on market efficiency.
- Hennig-Thurau et al. (2006) investigate to what extent two facets of employee emotions, namely service employees' display of positive emotions and the authenticity of their emotional labor, influence customers' assessments of service encounters. They use PLS to analyze the outcomes of an experiment. Bagozzi et al. (1991) show how PLS can be used to analyze marketing and consumer data obtained from experimental designs. Updated guidelines have been proposed by Streukens and Leroi-Werelds (2016).
- Ulaga and Eggert (2006) use PLS to develop and validate a higher-order construct "relationship value" in order to help firms differentiate in business relationships. This paper is also one example for the many papers in B2B marketing using PLS.
- In another B2B marketing paper, Smith and Barclay (1997) use PLS to analyze dyadic relationships to explore the role of trust in selling alliances.
- Johnson et al. (2006) use PLS to demonstrate that perceived value early in the customer life cycle affects loyalty intentions of mobile phone customers. Their paper is one of the relatively few who apply PLS to longitudinal data. Roemer (2016) offers a tutorial on how to analyze longitudinal data using PLS.

Marketing is not the only business research discipline that relies strongly on PLS as an analysis method. Neighboring disciplines, such as purchasing (Kaufmann and Gaeckler 2015), operations management (Peng and Lai 2012), and strategic management (Hair et al. 2012a; Hulland 1999), are also increasingly applying PLS path modeling.

The modularity of PLS path modeling, as introduced in the second section, opens up the possibility of replacing one or more steps with other approaches. For instance, the least squares estimators of the third step could be replaced with neural networks (Buckler and Hennig-Thurau 2008; Turkyilmaz et al. 2013). One could even replace the PLS algorithm in Step 1 with alternative indicator weight generators, such as principal component analysis (Tenenhaus 2008), generalized structured component analysis (Henseler 2012b; Hwang and Takane 2004), regularized generalized canonical correlation analysis (Tenenhaus and Tenenhaus 2011), or even plain sum scores. Since the iterative PLS algorithm would not serve as an eponym in these instances, one can no longer refer to PLS path modeling. However, it is still variance-based structural equation modeling.

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